Language Organization in Multilingual LLMs through SAE Feature Suppression

Anonymous Author(s)

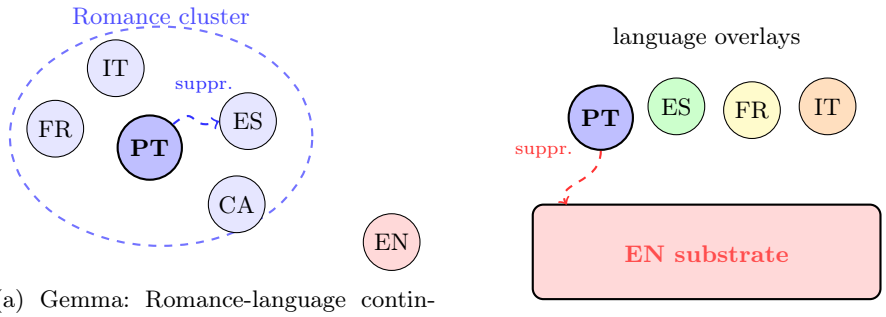
Anonymous Institution

Abstract. The way multilingual large language models organize the languages they speak depends on the model family, and we present causal evidence for this claim. Using Portuguese as a probe, we apply Sparse Autoencoder (SAE) feature suppression to Gemma-2-2B and Llama-3.1-8B: ablating the top Portuguese-specific features redirects Gemma to Spanish or Catalan and Llama to English. The two fallback directions expose internal geometries we name the *Romance-language continuum* (languages arranged by typological proximity, observed in Gemma) and the *English substrate* (non-English languages as overlays on an English default, observed in Llama). Portuguese is suited to surface this contrast because its Romance neighbors make any Romance fallback directly observable; a typologically isolated probe could not separate the continuum from the substrate. The contrast survives feature-agnostic controls: Gemma's Romance fallback requires PT-targeted suppression (random and EN-feature suppression preserve Portuguese), whereas Llama collapses to English under PT-targeted, EN-targeted, and L0-matched random suppression alike (Mann-Whitney $p = 0.69$ between PT and random), evidence that the substrate manifests as a generalized fragility. Probes in Spanish, Italian, and French confirm that Llama collapses to English on 42 of 45 Romance probes while Gemma preserves the source language. Pairwise masked FLORES cosine similarity (PT-ES > PT-EN at $p < 10^{-150}$ per layer in both models) and linear CKA at the terminal layer (PT-EN at 0.61 in Gemma vs. 0.74 in Llama) corroborate the divide. We also train 16 custom SAEs and find that the ReLU+L1 vs. TopK architecture ranking does not transfer across base models (TopK FVE 79.7% on Gemma against 98.0% on Llama). To our knowledge, this is the first mechanistic study to establish a causal cross-family contrast of language-organization paradigms.

Keywords: Mechanistic Interpretability · Sparse Autoencoders · Multilingual LLMs · Portuguese NLP · Feature Steering

1 Introduction

Modern large language models (LLMs) process dozens of languages simultaneously, yet how they organize those languages internally is not well understood. The literature offers two partial answers that have not been reconciled. Wendler et al. [20] argue that Llama models use English as an internal *lingua franca*,



(a) Gemma: Romance-language continuum

(b) Llama: English substrate

Fig. 1: Two architectures of multilingualism made visible through SAE feature-suppression. **(a)** In Gemma-2-2B, Portuguese sits inside a Romance cluster; suppressing PT-specific features redirects generation to the nearest typological neighbor (Spanish or Catalan). **(b)** In Llama-3.1-8B, non-English languages operate as overlays on an English substrate; suppressing PT-specific features strips the Portuguese overlay and reveals the English default.

with non-English inputs traversing English-centric intermediate representations. A separate line of work [13,19,7] documents shared multilingual spaces where languages cluster by typological similarity. Both pictures are backed by evidence, but the evidence comes from different models, methods, and analyses, leaving open whether multilingual LLMs share one organizational paradigm or several.

This paper offers causal evidence that both pictures are correct, each for a different model family. Using Portuguese as a probe in Sparse Autoencoder (SAE) feature-suppression experiments (Figure 1), we find that suppressing the top Portuguese-specific features redirects generation to Spanish or Catalan in Gemma-2-2B and to English in Llama-3.1-8B. The two fallback directions expose internal geometries we name the *Romance-language continuum*, in which languages are organized by typological proximity (observed in Gemma), and the *English substrate*, in which non-English languages behave as overlays on an English default (observed in Llama).

The choice of Portuguese is methodological rather than incidental. Its Romance neighbors (Spanish, Catalan, French, Italian) make any Romance fallback directly visible, so a continuation in Spanish or Catalan after suppression in Gemma reads off the model's next-nearest language in representational space; a typologically isolated probe with no close neighbors would conflate the continuum with the substrate. Portuguese has over 260 million speakers, with Brazilian Portuguese the dominant variety, and it remains under-represented in mechanistic interpretability work.

We make three contributions. First, we show that two distinct organizations of multilinguality coexist in current LLMs (the Romance-language continuum and the English substrate), and establish the contrast through causal feature suppression rather than correlational analysis. Second, the direction of genera-

tion collapse under suppression is itself a *lesion-based readout* of a language's representational neighborhood, and pairing it with a typologically embedded probe is what makes the paradigms distinguishable. Third, three complementary measurements corroborate the divide: raw-neuron monolinguality, pairwise FLORES similarity, and single-feature steering that surfaces a PT-EN bridge feature in Llama with no Gemma analogue.

2 Related Work

Sparse autoencoders decompose activations into overcomplete, sparse features that are more interpretable than raw neurons [5,1]. Gao et al. [8] scale SAE training and introduce the FVE and dead-feature metrics we adopt, and Shu et al. [17] survey the area; benchmarks [12,22] caution that proxy metrics do not always predict downstream utility, though SAEs remain well-suited to the feature-level analysis we pursue. We build on pretrained SAE suites for Gemma [14] and Llama [11] and extend feature-clamping [1] to language identity.

The multilingual-representation literature is split. One line argues that Llama-family models reason through English-centric intermediates: Wendler et al. [20] use logit-lens analyses on Llama 2, and Schut et al. [16] extend the picture with activation steering, reporting that English-derived steering vectors transfer better than those from other languages. A separate line [13,19,7] documents shared multilingual spaces where languages cluster by typological similarity, and the closest cross-family concurrent work (Brinkmann et al. [2], who train SAEs on Llama-3-8B and Aya-23-8B) shows that morphosyntactic concepts such as number, gender, and tense live on feature directions shared across typologically diverse languages. We read these pictures as compatible once paired with a model family: grammatical abstraction may be shared cross-lingually while language-identity geometry remains family-dependent. Our contribution turns that divide causal and family-contrastive: suppression redirects Gemma to a Romance neighbor and Llama to English, with Portuguese serving as the typologically embedded probe that makes the two paradigms distinguishable. Concurrent SAE-steering studies target typologically distant pairs within a single model (EN→ZH, EN→JA): LangFIR [21], SASFT [6], and Causal Language Control [3] (90% single-feature steering in Gemma-2B); our setting differs by being cross-family and Romance-focused, where shared representations make steering harder. On Portuguese itself, prior NLP work targets downstream performance [18]; we provide the first mechanistic catalog of Portuguese-specific features with causal validation.

3 Methodology

3.1 Models and Data

We analyze two model families (Table 1), chosen for their architectural differences and availability of pretrained SAEs.

Table 1: Base models and pretrained SAEs used in this study.

Model	Params	Layers	d_{model}	SAE d_{sae}	SAE Activation
Gemma-2-2B	2B	26	2,304	16,384	ReLU
Llama 3.1 8B	8B	32	4,096	32,768	JumpReLU

For multilingual evaluation we use FLORES-200 [4] parallel sentences in five languages (PT, EN, ES, FR, IT), 1,012 pairs per language; because the sentences are parallel translations, differential activation patterns reflect linguistic rather than semantic differences. SAE training data comes from Portuguese Wikipedia (∼2M tokens per model), tokenized with each model's native tokenizer (SentencePiece for Llama, BPE for Gemma); residual stream activations are extracted with TransformerLens [15] and stored as float16 tensors. We also analyze Gemma-2-9B (42 layers, d_{model}=3,584) for a within-family cross-scale comparison, using raw neuron activations since Gemma Scope SAEs were not available for the 9B model at the time of this study.

3.2 Sparse Autoencoder Architectures

We compare two SAE architectures, each trained on layer-10 residual stream activations from Portuguese Wikipedia:

ReLU+L1. The encoder maps an input activation $\mathbf{x} \in \mathbb{R}^d$ to a sparse code $\mathbf{f} = \text{ReLU}(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{pre}}) + \mathbf{b}_{\text{enc}}) \in \mathbb{R}^{d_{\text{sae}}}$, which the decoder reconstructs as $\hat{\mathbf{x}} = W_{\text{dec}}\mathbf{f}$. Training minimizes:

$$\mathcal{L}_{\text{ReLU}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda\|\mathbf{f}\|_1 \tag{1}$$

with decoder columns normalized to unit norm at each step.

TopK. Same encoder-decoder architecture, but enforces exact sparsity by retaining only the top-k encoder activations (applying ReLU to them) and zeroing the rest. The loss is MSE only:

$$\mathcal{L}_{\text{TopK}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \tag{2}$$

since sparsity (the ℓ_0-norm $\|\mathbf{f}\|_0 = k$ exactly) is enforced structurally rather than via penalty. We use L_0 throughout to refer to this ℓ_0-norm sparsity count.

For each base model we train 8 SAE configurations (totaling 16 custom SAEs): both architectures at $d_{\text{sae}} \in \{4{,}096, 16{,}384\}$, with $\lambda \in \{5 \times 10^{-4}, 10^{-4}\}$ for ReLU+L1 and $k \in \{32, 64\}$ for TopK. Training uses Adam (lr 3×10^{-4}) for 1 epoch with batch size 4,096 (Gemma) or 2,048 (Llama).

3.3 Differential Feature Analysis

To identify Portuguese-specific features, we compute a differential activation score. For each SAE feature i, we first compute the mean activation $\bar{a}_i^l$ across all FLORES tokens in language l. The differential activation score is:

$$s_i = \frac{\bar{a}_i^{\mathrm{PT}} - \frac{1}{|\mathcal{L}'|}\sum_{l\in\mathcal{L}'} \bar{a}_i^l}{\sigma_i} \tag{3}$$

where $\mathcal{L}' = \{\mathrm{EN, ES, FR, IT}\}$ is the set of contrast languages, $\mathcal{L} = \mathcal{L}' \cup \{\mathrm{PT}\}$ is the full set of five FLORES languages, and $\sigma_i = \mathrm{std}(\{\bar{a}_i^l\}_{l\in\mathcal{L}})$ normalizes by cross-language variability. Features with $s_i > 2$ are considered strongly Portuguese-specific: their Portuguese activation is more than 2 standard deviations above the cross-language mean.

We apply this analysis at 6 layers per model: layers 0, 5, 10, 15, 20, 25 for Gemma 2B; layers 0, 5, 10, 16, 21, 26 for Llama 8B (selected at approximately equivalent relative positions).

3.4 Monolinguality Metric

To compare how sharply languages separate within each model, we compute a monolinguality metric on raw residual-stream neurons (not SAE features). We operate on raw neurons so that the same procedure can be applied to Gemma-2-9B in the cross-scale analysis of Section 4.5, where no pretrained SAE suite is available. For each neuron i and language L, let $a_i(L) = |\overline{x_i^{(L)}}|$ denote the absolute mean activation of neuron i on FLORES sentences of language L. The inner mean is taken over content tokens only: padding (token 0 in Gemma; the eos token in Llama, which serves as pad) and the BOS token are excluded, then averaged across sentences. The masking step matters because FLORES sentences tokenize to different lengths across languages (Llama: PT 39.5 vs. EN 26.8 content tokens out of a 128-token window on average), and unmasked aggregation lets padding representations dominate the cross-language comparison. A neuron is *PT-dominant* at a given layer if $\arg\max_{L\in\mathcal{L}} a_i(L) = \mathrm{PT}$, i.e., its absolute activation is largest on Portuguese among the five FLORES languages. For each PT-dominant neuron, its monolinguality is

$$m_i(\mathrm{PT}) = \frac{a_i(\mathrm{PT})}{\sum_{L\in\mathcal{L}} a_i(L)}, \qquad \mathcal{L} = \{\mathrm{PT, EN, ES, FR, IT}\}. \tag{4}$$

Values close to $1/|\mathcal{L}| = 0.2$ indicate a neuron that activates comparably across all languages; values near 1 indicate a neuron that fires almost exclusively on Portuguese. We report, per model, the mean of $m_i(\mathrm{PT})$ over PT-dominant neurons at each analyzed layer, and a single per-model figure obtained by averaging across those layers. For raw-neuron analyses (monolinguality and the pairwise similarity of Section 5) we use the layers listed in Section 3.3 and add Llama's terminal layer 31, so the per-model average is taken over 6 layers for Gemma and 7 for Llama, placing each model's deepest analyzed layer at 100% relative depth.

Table 2: Best SAE per architecture per model (evaluated on FLORES PT activations, layer 10). FVE = Fraction of Variance Explained. L_0 = mean number of non-zero features per activation (ℓ_0-norm). Dead = fraction of features never activated.

Model	Architecture	FVE (%)	L_0	MSE	Dead (%)
Gemma 2B	ReLU+L1 (d=4k, λ=1e-4)	99.19	4,040	0.063	0.0
	TopK (d=16k, k=64)	79.65	64	1.578	35.3
Llama 8B	ReLU+L1 (d=16k, λ=1e-4)	**99.83**	7,684	0.001	1.1
	TopK (d=16k, k=64)	98.01	64	0.009	31.3

3.5 Steering Experiments

We perform causal interventions by hooking into the model's forward pass at layer 10 and modifying SAE-encoded features. For amplification, we encode the residual activation through the SAE, multiply the top-20 Portuguese-specific features by factor $\alpha \in \{1, 2, 5, 10\}$, decode back, and replace the original activation; this tests whether increasing PT feature magnitudes forces Portuguese generation. For suppression, the same procedure is applied but the top-20 features are set to zero, testing whether removing PT features causes language switching and, in particular, which language the model defaults to.

For Gemma 2B we use Gemma Scope (layer 10, 16k features); for Llama 8B we use Llama-Scope (layer 10, 32k features, JumpReLU). We test on 10 prompts per language (PT, EN, ES, FR), generating 50 tokens with temperature 0.7.

4 Experiments and Results

4.1 Reconstruction Quality

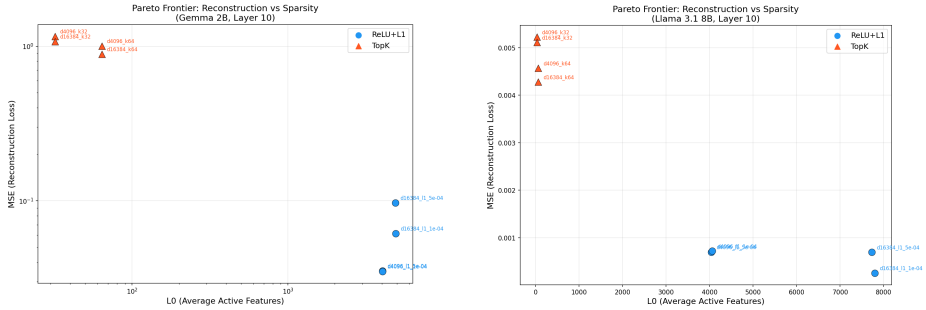
Table 2 reports the best configuration per architecture for each model; Figure 2 shows the full Pareto frontier across all 16 custom SAEs.

TopK performance is model-dependent: 79.7% FVE on Gemma 2B (35.3% dead) versus 98.0% on Llama 8B (31.3% dead). ReLU+L1 exceeds 99% FVE on both models with near-zero dead rates, making it the more robust choice; Section 5.2 discusses the likely source of the gap.

4.2 Language Probing

Linear probing (logistic regression, 5-fold CV) on mean-pooled layer-10 activations for 5-way language classification yields near-perfect accuracy for both models (Table 3), confirming that language identity is linearly separable in the residual stream and that SAE encoding preserves this information.

All accuracies exceed 99.8%, meaning a simple linear classifier can distinguish languages from a single layer's activations. SAE features achieve comparable



(a) Gemma 2B (b) Llama 3.1 8B

Fig. 2: Pareto frontier: reconstruction loss (MSE) vs. sparsity (ℓ_0-norm L_0) for all 8 SAE configurations per model. Blue circles = ReLU+L1; orange triangles = TopK. Gemma axes are on a log scale; Llama axes are linear.

Table 3: Language classification accuracy (5-fold CV, 1,012 samples per language). The best custom SAE (by FVE) is used for SAE feature probing.

Representation	**Gemma 2B**	**Llama 8B**
Raw activations (d_{model})	99.94% ± 0.12	99.98% ± 0.04
SAE features (d_{sae})	99.98% ± 0.04	99.88% ± 0.15

accuracy to raw activations despite imperfect reconstruction (FVE of 99.2% for the Gemma SAE and 99.8% for the Llama SAE used for probing), confirming that the SAE decomposition retains language-relevant information.

4.3 Portuguese-Specific Features

Differential analysis (Eq. 3) with pretrained SAEs identifies strongly Portuguese-specific features ($s_i > 2$) across all analyzed layers in both models (Table 4).

Both models contain thousands of strongly PT-specific features. Normalizing for SAE dictionary size: Gemma has 4,482/16,384 = 27.4% and Llama has 3,811/32,768 = 11.6%. Gemma 2B concentrates features at layer 0 (1,134) and recovers a second peak at layer 25 (777); Llama 8B peaks at layer 10 (747) and declines to 452 at layer 26. The padding-masked Gemma differential roughly doubles the counts reported in earlier drafts that used unmasked aggregation, while preserving the qualitative pattern.

From the set of strongly PT-specific features in Gemma 2B we selected the top 20 features per layer by differential score, yielding 120 features across the 6 analyzed layers. We categorize each feature by its 10 top-activating content tokens in FLORES PT using a rule-based classifier that checks for function words, PT-typical suffix patterns, capitalized proper nouns, punctuation, and whitespace (Table 5).

Table 4: Strongly PT-specific feature counts ($s > 2$) at each analyzed layer using pretrained SAEs (column headers indicate layer indices; Gemma 2B: layers 0, 5, 10, 15, 20, 25; Llama 8B: layers 0, 5, 10, 16, 21, 26). Both rows use padding-masked analysis: tokens identified as padding or BOS are excluded before computing per-feature mean activations. The differential activation score has a theoretical ceiling of $5/2 = 2.5$ (when a feature activates only on PT among the five languages, its score equals mean/std = 2.5).

Model (SAE)	**Layer 0**	**Layer 5**	**Layer 10**	**Layer 15/16**	**Layer 20/21**	**Layer 25/26**	**Total**
Gemma 2B (16k)	1,134	835	733	473	530	777	4,482
Llama 8B (32k)	698	675	747	646	593	452	3,811

Table 5: Categories of Portuguese-specific features in Gemma 2B (120 features across 6 layers, classified by their top-10 activating content tokens). Counts assigned by a deterministic rule-based classifier; the residual "Mixed" category covers features whose top tokens span more than one of the listed classes.

Category	**L0**	**L5**	**L10**	**L15**	**L20**	**L25**	**Total**	**Example tokens**
Lexical (content)	3	10	11	9	5	6	44	"Sul", "modern", "cit"
Syntactic (function)	15	3	2	2	4	5	31	"de", "do", "os", "uma"
Mixed	1	1	3	2	2	5	14	heterogeneous
Lexical (proper noun)	0	3	2	1	4	0	10	"Brasil", "Portugal"
Punctuation	0	1	1	4	2	2	10	".", ",", "("
Subword fragment	1	2	1	2	3	1	10	"rália", "-icos"
Whitespace	0	0	0	0	0	1	1	" "

A layer-wise gradient emerges: layer 0 is dominated by syntactic features (15 of 20 features are function words: "de", "do", "os", "uma"), middle layers (5–15) shift toward lexical content words and proper nouns, and later layers retain a mix of syntactic, punctuation, and content features. The original draft of this work classified 68 of 120 features as "Unclassified" due to noise on padding positions; padding-masked analysis reduces that residual to 14.

For Llama 8B, many top features initially activated on padding tokens, an artifact of differential tokenization lengths (PT averages 40.5 tokens/sentence vs. EN's 27.8). Masking padding positions before computing mean activations reduces the strongly PT-specific count from 4,143 to 3,811 (-8%); Table 6 shows the corrected top-20.

We restricted manual inspection to the top-20 features at the SAE-training layer (layer 10) for Llama, against 120 features across all 6 analyzed layers for Gemma, reflecting the cost of token-level inspection of Llama-Scope's larger dictionary. After masking, syntactic features (prepositions "de"/"com", articles "a"/"os"/"um") dominate, with morphological features ("são", "â", suffixes "-icos") also present, mirroring Gemma's layer-10 composition. Methodologically, differential analysis on padded sequences can produce spurious results when languages

Table 6: Top-20 Llama-Scope PT features (layer 10) after padding masking. Features are categorized by their top-activating tokens on genuine (non-padding) Portuguese text.

Category	Count	Example tokens	Note
Syntactic	14	“de”, “os”, “a”, “um”, “que”, “com”	Prepositions, articles, conjunctions
Morphological	4	“são”, “â”, “-icos”, “-ode”	PT verb forms, suffixes
Mixed/subword	2	“ab”, “di”	Subword fragments in PT contexts

tokenize to different lengths, a pitfall we have not seen documented in the SAE interpretability literature.

4.4 Steering Results

Table 7 delivers our central result: suppressing the top-20 PT-specific features redirects generation in family-specific directions, and that direction distinguishes the two paradigms of Figure 1.

In Gemma 2B, suppression consistently lands on a closely related Romance language: Spanish for most prompts, and Catalan for *“A capital do Brasil é”*. The Catalan output is informative because Catalan is not among the five FLORES evaluation languages, so the fallback is not a ranking over the measured set but reflects a Romance geometry that includes unevaluated varieties. For *“A língua portuguesa é falada em”* the Spanish continuation also preserves topical coherence: the model keeps talking about languages, just in another Romance one. Llama 8B, in contrast, falls back to English on every suppression prompt, consistent with the English-pivot picture of Wendler et al. [20]: non-English surface forms behave as overlays on an English-centric base, so stripping the Portuguese overlay reveals it.

Two controls test whether the fallback direction is driven by PT-feature targeting or by any feature-level perturbation: an L0-matched random condition (20 features matched in PT activation rate but with $|s| \leq 1$, three random seeds) and the top-20 EN-specific features. Across 45 generations per random condition and 15 per EN condition (5 prompts $\times$ 3 generation seeds), Gemma preserves Portuguese under both controls (random: 36/45; EN: 13/15) while only PT-feature suppression breaks Portuguese (11/15 shift to a Romance language). Llama collapses to English at the same rate under PT, EN, and random conditions (10/15, 10/15, 31/45 respectively; Mann-Whitney $p = 0.69$ for PT vs. random). Extending the protocol to Spanish, Italian, and French probes, where we suppress the top-20 features specific to each language on prompts in that language, Llama collapses to English on 42 of 45 Romance probes (ES 14/15, IT 15/15, FR 13/15) while Gemma preserves the source language (ES 12/15, IT 15/15, FR 15/15). Gemma’s Romance-language collapse is therefore feature-specific, while Llama’s English collapse reflects a generalized fragility of language identity at layer 10 that generalizes to every Romance overlay we tested.

Table 7: Steering examples. Baseline: normal generation; Suppression: top-20 PT features set to zero; Amplification: top-20 PT features multiplied by 10. Fallback-language labels (**in bold**) were assigned by manual inspection of the generated text.

Prompt	Mode	Gemma 2B output	Llama 8B output
A capital do Brasil é	Base	palco de diversos eventos e festivais...	o centro mais importante do país...
	Suppr.	la una metròpolis ambduna de les fa uns centenars... **(Catalan)**	a capital, 1 capital, is 1 capital... **(English)**
	Amplif.	uma das métroas mais importantes da metropode... **(PT, degraded)**	que que que u u u... **(collapsed)**
O rio mais longo do mundo é	Suppr.	el férriable del continente... **(Spanish)**	the longest the most long the... **(English)**
	Amplif.	o Brasil, e sua extensão é de 235 mil quilômetros... **(PT)**	mais do mais mais... **(collapsed)**
A língua portuguesa é falada em	Base	22 países, sendo Portugal um dos principais...	mais de 70 países, cerca de 200 milhões de falantes...
	Suppr.	todos os continentes, enrapapeando la lengua española... **(Spanish)**	il...?...?...? **(degenerate)**

Amplification exposes a complementary asymmetry. Multiplying the top-20 PT features by 10 leaves 18 of 25 Gemma generations (5 prompts $\times$ 5 seeds) in Portuguese against only 6 of 25 in Llama, with 14 of 25 Llama outputs collapsing to English. Continuation length drops in both models (Gemma 24% [bootstrap 95% CI 11–37], Llama 16% [0–31] at factor 10), and the Mann-Whitney test does not separate the two ($p = 0.69$); the family contrast is in the language identity of the surviving output, not in the magnitude of the collapse. The most likely mechanistic explanation is sparsity: Llama-Scope features fire on roughly $L_0 = 24$ feature dimensions per content token vs. $L_0 = 86$ for Gemma Scope, so 20-feature interventions perturb 83% of the active code in Llama against 24% in Gemma. Direct magnitude comparison between the two SAEs is confounded by different activation functions (ReLU+L1 vs. JumpReLU) and input normalization, so we report sparsity as the architecture-independent quantity that supports this account.

Amplifying individual Llama-Scope features (factor 5, prompt *"A capital do Brasil é"*) exposes distinguishable specializations. Feature 16811 produces repetitions drawn exclusively from {Portugal, Portuguese, Brazil}, a "Lusophone-world" concept; feature 18397 generates a PT-EN hybrid (*"a capital do Brazil is the capital of Britain countries"*), a cross-lingual bridge that surfaces the sub-

Table 8: Cross-scale comparison within the Gemma family. Cosine similarity between masked-mean PT and ES/EN/FR activations and the count of strongly PT-specific neurons ($s > 2$, applied to raw neurons rather than SAE features). All entries use the same padding-masked aggregation as Section 3.4.

9B Layer	**Rel. pos.**	**cos(PT,ES)**	**cos(PT,FR)**	**cos(PT,EN)**	$n_{s>2}$
0	0%	0.959	0.944	0.887	234
8	20%	0.982	0.977	0.957	302
16	39%	0.996	0.995	0.988	183
24	59%	0.994	0.991	0.986	252
32	78%	0.982	0.960	0.962	232
41	100%	0.640	0.519	0.461	402

strate at the feature level; feature 16096 yields Brazil-anchored repetitions (*"BC Brazil is BC Brazil is is is..."*). The bridge feature has no direct Gemma analogue in our inspected inventory, matching a picture in which Gemma's Romance languages share representational space directly rather than routing through English.

Progressive steering on Gemma 2B localizes where language identity lives. Amplifying the top-20 PT-specific features (factor 5) at an increasing set of layers and scoring continuations with a PT-likeness heuristic $0.3\,c + 10\,w$ (Portuguese-specific characters c and function-word ratio w), layer 25 alone yields 4.5, layers 15+20+25 together peak at 7.2, and adding layer 10 drops the score to 0.9 (incoherent text). The heuristic agrees with fastText LID on the coherent subset (Spearman $\rho = 0.83$, $n = 55$); published LIDs collapse on the degenerate outputs of this experiment, so marker counting is the appropriate readout. Language identity therefore lives in mid-to-late layers (15–25), and steering early layers disrupts foundational representations that later layers depend on, which also explains why multi-layer suppression succeeds where single-layer steering does not.

4.5 Cross-Scale Analysis

To disentangle family effects from scale effects, we also analyzed Gemma-2-9B (42 layers, $d_{\text{model}} = 3{,}584$) using raw neuron activations (Table 8).

The 9B model shows progressive linguistic differentiation (Figure 3): cosine similarity between Portuguese and Spanish masked-mean activations decreases from 0.959 at layer 0 to 0.640 at layer 41, with the sharpest drop in the final third. The PT-EN drop is steeper still (0.887 $\rightarrow$ 0.461) and a clean Iberian ordering emerges at the terminal layer (ES > FR > EN: 0.640, 0.519, 0.461). The number of strongly PT-specific neurons rises from 234 at layer 0 to 402 at layer 41. The 2B model, by contrast, shows a flatter trajectory at the analyzed layers under the same masked aggregation. We rely on the qualitative contrast (flat vs progressive) rather than absolute magnitudes, since the 2B and 9B numbers live in different spaces: 2B scores in Section 4.3 use Gemma Scope SAE features,

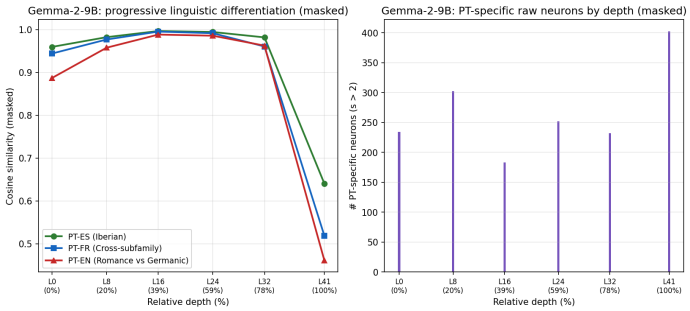


Fig. 3: Progressive linguistic differentiation in Gemma-2-9B: cosine similarity between Portuguese and Spanish/English mean activations decreases in deeper layers, while the number of PT-specific neurons increases. This gradient is absent in the 2B model.

while 9B scores here use raw neuron activations because no pretrained SAE suite was available for the 9B model at the time of this study. The pattern suggests that larger models develop finer language separation in deeper layers, a property not visible at the 2B scale. The cosine values reported here differ from earlier drafts of this work, which aggregated over all token positions including padding and BOS; the corrected (masked) cosines are systematically lower because shared padding representations no longer inflate cross-language similarity.

5 Discussion

5.1 Cross-Family Language Organization

A pairwise cosine similarity analysis on padding-masked FLORES activations supports a layered typological separation in both models: PT-EN distance (Romance vs. Germanic) is already the largest at the earliest layers, PT-FR distance (cross-subfamily) grows through middle layers, and PT-ES distance (Iberian) diverges only at the deepest analyzed layer. For Gemma 2B at layer 25 the masked centroids give cos(PT,ES)= 0.55, cos(PT,FR)= 0.38, and cos(PT,EN)= 0.18; for Llama 8B at layer 31 the same ordering holds at higher absolute values: 0.72, 0.62, 0.42. The PT-ES vs. PT-EN gap is reproducible at every layer in both models when computed per sentence (paired Wilcoxon, $p < 10^{-150}$ at every layer with $N = 1012$ sentences). Linear CKA between PT and other-language sentence representations at the terminal layer reproduces this ordering and quantifies the cross-model contrast: PT-ES CKA is 0.80 in both Gemma and Llama, while PT-EN CKA is 0.61 in Gemma against 0.74 in Llama, a 0.13 gap that is the single strongest quantitative signature of the English-substrate paradigm in our data.

Beyond Table 7, the identity and texture of the fallback strengthen the picture. For Gemma, *"A língua portuguesa é falada em"* continues in Spanish (*"todos*

os continentes, enrapapeando la lengua española..."), preserving the topic, and another prompt drifts into Catalan (*"la una metròpolis ambduna de les fa uns centenars"*); since Catalan is absent from our five evaluation languages, the continuum reaches beyond the set we measure. Gemma 2's training distribution [9] plausibly encourages this shared Romance structure, although Google does not publicly detail its language composition. Llama, in contrast, falls back to English (*"the longest the most long the is the more long the"* from *"O rio mais longo do mundo é"*), consistent with Wendler et al. [20] and with Meta's predominantly English pretraining mix for Llama 3 (only 8% multilingual tokens in the final data mix) [10], which would make English a natural base layer.

The padding-masked monolinguality metric of Eq. 4 does not by itself separate the two models: averaged across analyzed layers, Llama has PT monolinguality 0.367 (95% CI 0.348–0.386) against Gemma's 0.363 (0.336–0.391), a paired difference of 0.004 over relative-depth-matched layers (Wilcoxon paired $p = 0.28$). The substrate paradigm is supported instead by the suppression and probe experiments of Section 4.4 and by the CKA gap reported above. Among individual features, amplifying Llama-Scope feature 18397 produces a PT-EN hybrid (*"a capital do Brazil is the capital of Britain countries"*) with no analogue in our inspected Gemma inventory, consistent with a Llama representation in which Romance overlays route through English rather than sharing space with one another. These suppression experiments may also be read as intentionally induced code-switching, connecting to SASFT [6].

The two paradigms suggest different affordances: a Romance continuum may permit finer-grained language steering by "sliding" between related languages, while an English substrate raises concerns about English-centric biases propagating to all other languages.

5.2 TopK Performance Across Models

The TopK gap (79.7% FVE on Gemma 2B vs. 98.0% on Llama 8B) appears under identical hyperparameters. A plausible explanation is the intrinsic dimensionality of the activations: Llama's d_{model}=4,096 (nearly twice Gemma's 2,304) offers a higher-dimensional space where activations may concentrate on a k-sparse basis more readily, while Gemma's lower-dimensional activations diffuse across many components and defeat exact-k sparsity. ReLU+L1 adapts its sparsity per sample and exceeds 99% FVE on both models; TopK produces 31% dead features on Llama and 35% on Gemma at the best configuration, versus <1.2% for ReLU+L1. The practical takeaway is that TopK transfer is not automatic: practitioners should validate on their specific base model.

5.3 Limitations

Our two models differ simultaneously in family, size, attention variant, and training mix, so the observed differences in fallback behavior may reflect any combination of these factors. Scale alone is unlikely to explain the paradigm: Gemma-2-9B (larger than Llama-3.1-8B in parameter count) preserves a Romance-continuum

organization through its terminal layer (cos(PT,ES)=0.640 vs. cos(PT,EN)=0.461, Table 8) rather than drifting toward an English-centric geometry. A fully controlled comparison would still require same-size models from different families. The pretrained SAEs themselves differ in activation function, dictionary size, and training procedure, so we rely on qualitative contrasts and on the architecture-independent sparsity statistic (L_0) rather than on absolute magnitudes. An earlier draft of this work aggregated FLORES activations without masking padding and BOS positions; the corrected (masked) numbers reduce the apparent Llama vs. Gemma gap in monolinguality and amplify the typological-continuum signal in pairwise cosines, and we report only the masked values throughout. Language identification of generated continuations uses fastText LID-176 top-1 with no confidence threshold, so degenerate generations may be misclassified; bootstrap confidence intervals use a fixed set of five Portuguese prompts and three to five generation seeds, and generalization to other prompts is not directly tested. Steering generation uses T=0.7, so the specific strings in Table 7 are illustrative; the qualitative conclusions (fallback identity, feature categories) are stable across prompts and runs. Recent work [3] shows 90% single-feature steering between typologically distant languages, indicating that the difficulty we observe within Romance languages reflects genuinely shared representations rather than a methodological wall.

In scope, we analyze 6 of 26–32 layers per model, 5 European languages (4 Romance), and a single corpus (FLORES-200); the European focus may inflate Portuguese specificity scores and other genres may activate different features. 57% of Gemma's 120 inspected features remain unclassified; automated LLM-based interpretation could improve coverage but introduces its own biases. Future work should extend to additional model families (Mistral, Qwen) and to same-size comparisons, include typologically diverse languages, and investigate PT-BR vs. PT-PT dialect variation.

6 Conclusion

We presented causal evidence for two architectures of multilingualism. Suppressing Portuguese-specific features redirects Gemma-2-2B to Spanish or Catalan and Llama-3.1-8B to English, exposing internal geometries we name the Romance-language continuum and the English substrate. The contrast survives an L0-matched random-feature control (Gemma's Romance fallback requires PT-targeted suppression; Llama collapses to English under any matched 20-feature perturbation) and generalizes to Spanish, Italian, and French probes (Llama 42 of 45 to English; Gemma preserves the source language). Pairwise masked FLORES cosine similarity, linear CKA at the terminal layer (PT-EN gap of 0.13 between Gemma and Llama), and single-feature steering that surfaces a PT-EN bridge feature in Llama with no Gemma analogue corroborate the divide. Portuguese is what makes the paradigms distinguishable: a typologically embedded probe reveals the continuum where an isolated probe could not. Contrasting two families rather than describing one suggests that training data composi-

tion shapes the organizational paradigm itself, a hypothesis further cross-family studies can test.

Reproducibility. Source code, configurations, per-experiment result files, and the figures reported here are available at https://anonymous.4open.science/r/llm-pt-interpretability-EC0D/[1].

References

1. Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., et al.: Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread (2023)
2. Brinkmann, J., Wendler, C., Bartelt, C., Mueller, A.: Large language models share representations of latent grammatical concepts across typologically diverse languages. arXiv preprint arXiv:2501.06346 (2025)
3. Chou, C.T., Liu, G., Sun, J., Blondin, C., Zhu, K., Sharma, V., O'Brien, S.: Causal language control in multilingual transformers via sparse feature steering. arXiv preprint arXiv:2507.13410 (2025)
4. Costa-jussà, M.R., Cross, J., Çelebi, O., Elbayad, M., Heafield, K., Heffernan, K., Kalbassi, E., Lam, J., Licht, D., Maillard, J., et al.: No language left behind: Scaling human-centered machine translation. arXiv preprint arXiv:2207.04672 (2022)
5. Cunningham, H., Ewart, A., Riggs, L., Huben, R., Sharkey, L.: Sparse autoencoders find highly interpretable features in language models. arXiv preprint arXiv:2309.08600 (2023)
6. Deng, B., Wan, Y., Yang, B., Huang, F., Wang, W., Feng, F.: SASFT: Sparse autoencoder-guided supervised finetuning to mitigate unexpected code-switching in LLMs. In: Proceedings of the International Conference on Learning Representations (ICLR) (2026)
7. Deng, B., Wan, Y., Yang, B., Zhang, Y., Feng, F.: Unveiling language-specific features in large language models via sparse autoencoders. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Association for Computational Linguistics (2025)
8. Gao, L., la Tour, T.D., Tillman, H., Goh, G., Troll, R., Radford, A., Sutskever, I., Leike, J., Wu, J.: Scaling and evaluating sparse autoencoders. arXiv preprint arXiv:2406.04093 (2024)
9. Gemma Team: Gemma 2: Improving open language models at a practical size. arXiv preprint arXiv:2408.00118 (2024)
10. Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A., et al.: The Llama 3 herd of models. arXiv preprint arXiv:2407.21783 (2024)
11. He, Z., Shu, W., Ge, X., et al.: Llama scope: Extracting millions of features from llama-3.1-8b with sparse autoencoders. arXiv preprint arXiv:2410.20526 (2024)
12. Karvonen, A., Rager, C., Lin, J., Tigges, C., Bloom, J.I., Chanin, D., Lau, Y.T., Farrell, E., McDougall, C.S., Ayonrinde, K., Till, D., Wearden, M., Conmy, A.,

[1] Anonymous for double-blind review; released non-anonymously upon acceptance. SAE checkpoints and cached activations are not redistributed due to size; scripts regenerate them from public base models and FLORES-200 / Portuguese Wikipedia.

Marks, S., Nanda, N.: SAEBench: A comprehensive benchmark for sparse autoencoders in language model interpretability. In: Forty-second International Conference on Machine Learning (2025), https://openreview.net/forum?id=qrU3yNfX0d
13. Kojima, T., Gu, J., Reid, M., Matsuo, Y., Iwasawa, Y.: Multilingual large language models share semantic representations across languages. In: Findings of ACL (2024)
14. Lieberum, T., Rajamanoharan, S., Conmy, A., Smith, L., Varma, N., Kramár, J., Bolukbasi, T., Velez-Grau, A., et al.: Gemma scope: Open sparse autoencoders everywhere all at once on gemma 2. arXiv preprint arXiv:2408.05147 (2024)
15. Nanda, N., Bloom, J.: Transformerlens. GitHub repository (2022), https://github.com/TransformerLensOrg/TransformerLens
16. Schut, L., Gal, Y., Farquhar, S.: Do multilingual LLMs think in English? arXiv preprint arXiv:2502.15603 (2025)
17. Shu, D., Wu, X., Zhao, H., Rai, D., Yao, Z., Liu, N., Du, M.: A survey on sparse autoencoders: Interpreting the internal mechanisms of large language models. Findings of EMNLP (2025)
18. Souza, F., Nogueira, R., Lotufo, R.: BERTimbau: Pretrained BERT models for Brazilian Portuguese. In: Intelligent Systems (BRACIS). pp. 403–417. Springer (2020)
19. Tang, T., Luo, W., Huang, H., Zhang, D., Wang, X., Zhao, X., Wei, F., Wen, J.R.: Language-specific neurons: The key to multilingual capabilities in large language models. arXiv preprint arXiv:2402.16438 (2024)
20. Wendler, C., Veselovsky, V., Monea, G., West, R.: Do llamas work in english? on the latent language of multilingual transformers. arXiv preprint arXiv:2402.10588 (2024)
21. Wong, S.H., Sajjad, H., Siddique, A.: Langfir: Discovering sparse language-specific features from monolingual data for language steering. arXiv preprint arXiv:2604.03532 (2026)
22. Wu, Z., Arora, A., Geiger, A., et al.: AxBench: Steering LLMs? even simple baselines outperform sparse autoencoders. In: Proceedings of the 42nd International Conference on Machine Learning (ICML). PMLR (2025)